Towards an early detection and improved management of schistosomiasis in Africa through AI and the One Health approach

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Abstract

Schistosomiasis remains a major public health issue, particularly in sub-Saharan Africa. Despite control efforts, challenges persist due to complex interactions between human, animal, and environmental factors. This PRISMA-based review (2011–2025) explores the potential of artificial intelligence and the One Health approach to improve disease detection and management. However, issues such as limited dataset diversity, regional adaptability, and integration into health systems remain.

Introduction

Schistosomiasis remains a major neglected disease in sub-Saharan Africa. Current diagnostic and control strategies are limited, particularly in low-prevalence settings. AI and predictive modeling offer new opportunities, but existing approaches are often unimodal. Integrating AI within a One Health framework could improve diagnosis, risk prediction, and control strategies.

Comprehensive framework for data profiling

This study follows the PRISMA framework to review AI applications within a One Health approach for schistosomiasis. A search (2011–2025) across major databases identified 479 records, with 102 analyzed and 26 selected for indepth review. Results highlight the complementarity of AI and One Health in improving detection, prediction, and disease management.

Literature Review

Tallam et al. [1] and Ward et al. show that deep learning models (CNN, R-FCN) achieve high accuracy (up to 99.6%) for detecting schistosomiasis from images. These results highlight the potential of AI for automated diagnosis, although limitations remain in terms of clinical validation and real-world deployment.

Wood et al. and Andrade-Pacheco et al. use statistical and spatial sampling approaches to predict prevalence and detect hotspots. Environmental factors such as aquatic vegetation and human-water contact are key predictors, but limitations include lack of field validation and incomplete geospatial data.

Liu et al. and Koudenoukpo et al. apply AI methods (U-Net, SOM, Random Forest) to map risk areas using satellite and ecological data. While these approaches identify important environmental drivers, their generalization is limited by geographic specificity and insufficient seasonal data.

Rohr et al. and Singer et al. combine ecological interventions and machine learning for prevention. Vegetation removal significantly reduces infection risk, and predictive models help identify hotspots. However, these methods are limited by missing key data (e.g., snail populations) and limited transferability.

Comparative Analysis and Identified Research Gaps

Recent studies show that AI improves schistosomiasis diagnosis, prevalence prediction, and risk mapping. Deep learning models (VGG16, U-Net, ResNet) achieve high performance, while environmental models help identify hotspots. However, most approaches remain unimodal and lack strong clinical and field validation. Data limitations and poor interoperability reduce model generalization. Future research should focus on multimodal integration and better consideration of temporal and socio-behavioral factors.

First results of our study

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df.head(n=14)
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	site d'etude	vegetation	ph	Cat PH	conductivité	cat conductivité	temperature	TDS	Colonne1	Salinité	...	animaux	plastique	mollus
0	kothiedy	OUI	9.54	alcalin	4.15	faible	32.1	16.98	excellent	1.98	...	OUI	NON	
1	kothiedia	NON	10.00	alcalin	16.98	faible	31.8	1.43	excellent	8.43	...	OUI	NON	
2	ndoyenne	OUI	7.70	neutre	191.00	normale	25.3	191.00	excellent	0.97	...	OUI	OUI	
3	boguene goli	OUI	6.97	neutre	2.12	faible	29.0	1.56	excellent	1.11	...	OUI	NON	
4	louboudou	OUI	7.05	neutre	143.90	normale	26.4	1.02	excellent	0.67	...	OUI	NON	
5	inconnu	OUI	7.86	alcalin	1342.00	normale	25.0	904.00	bon	0.66	...	OUI	NON	

Figure 1: Overview of ecological analysis data

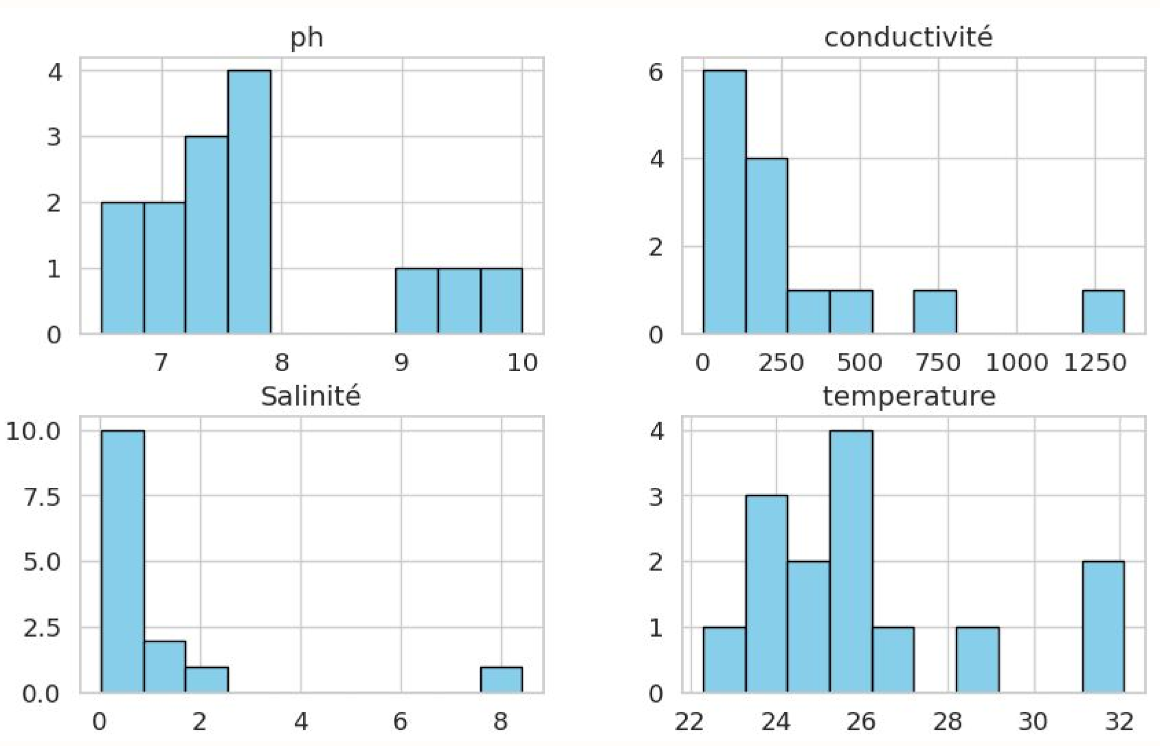

Figure 2: Distribution of physicochemical parameters

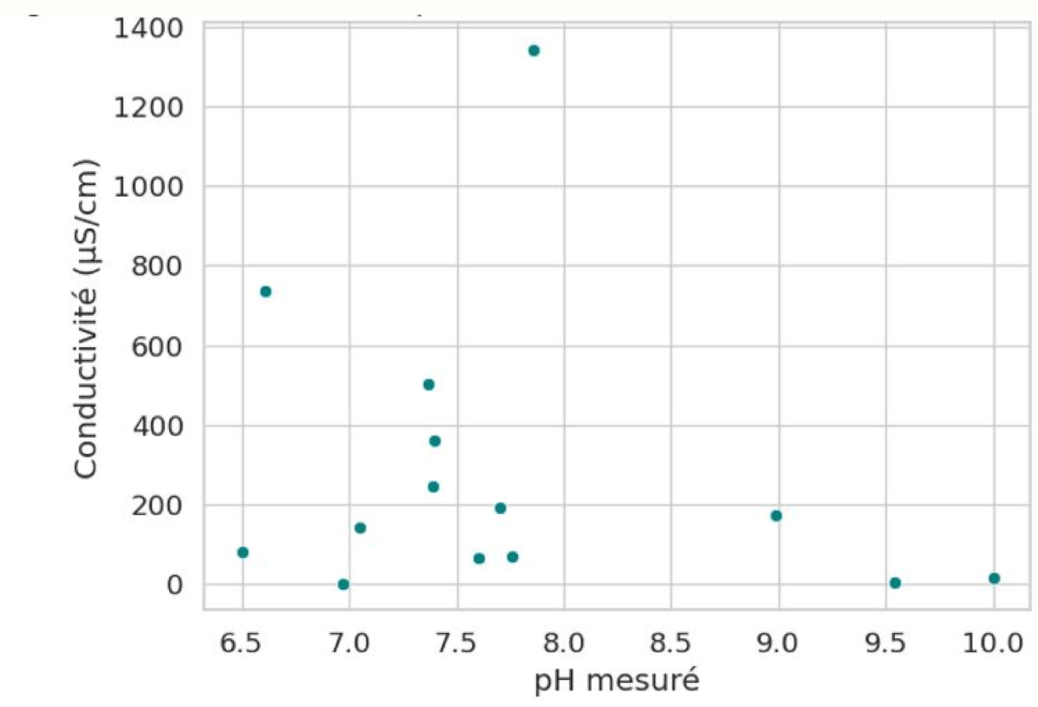

Figure 3: Relationship between pH and conductivity

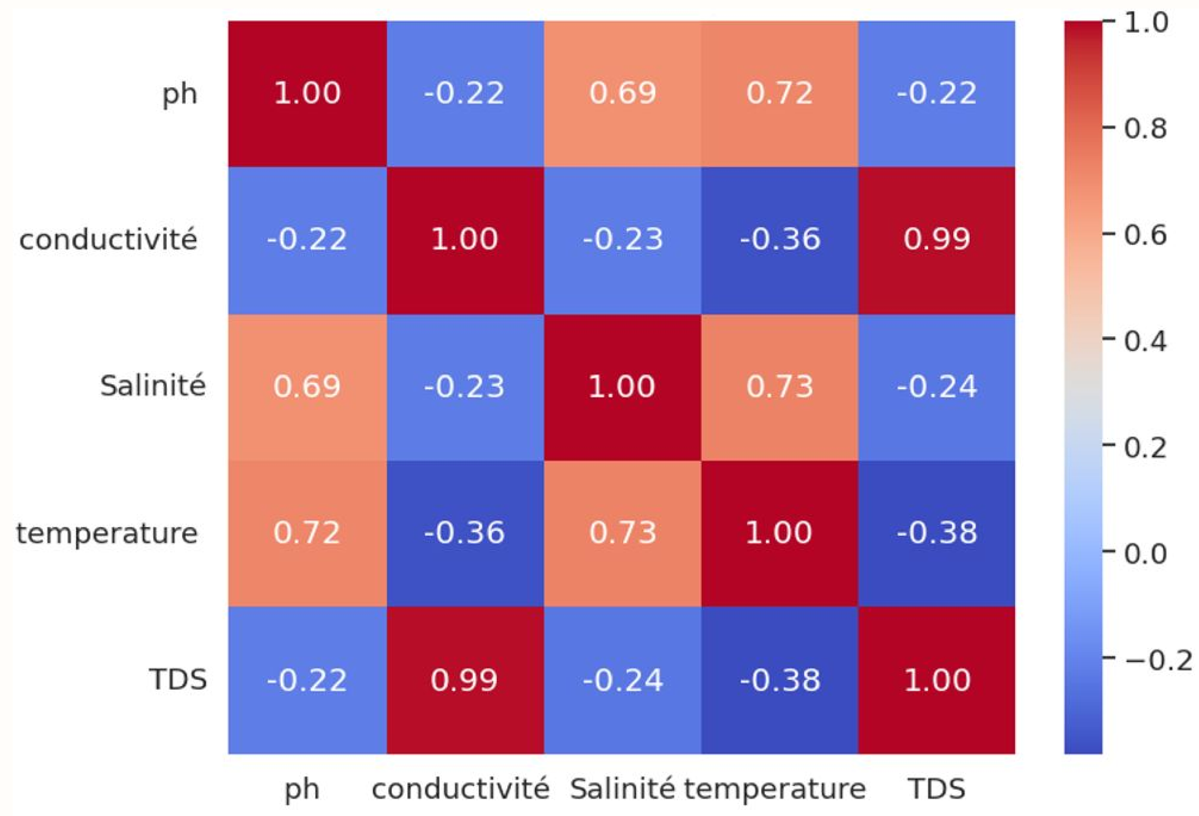

Figure 4: Correlation matrix between environmental parameters

The results indicate a warm, slightly alkaline freshwater environment (pH 8–8.5) with low salinity. Conductivity and TDS show spatial variability, while pH and conductivity are not correlated. A strong correlation between conductivity and TDS (0.99) suggests they reflect the same property, whereas other parameters form a separate group.

CONCLUSION

This study reviews the use of artificial intelligence within the One Health framework to combat schistosomiasis in Africa. It highlights the disease's complexity due to interactions between human, animal, and environmental factors. Existing research shows progress in diagnosis, prediction, and risk mapping using AI. However, there is no multimodal model integrating diverse data, and datasets remain fragmented and limited. Future work aims to develop a robust multimodal predictive model for improved decision support.

References

[1] K. Tallam, Z. Y.-C. Liu, A. Chamberlin, I. Jones, P. Shome, G. Riveau, R. Ndione, L. Bandagny, N. Jouanard, P. Eck, T. Ngo, S. Sokolow, and G. De Leo, "Identification of snails and schistosoma of medical importance via convolutional neural networks: A proof-of-concept application for human schistosomiasis," *Frontiers in Public Health*, vol. 9, p. 642895, 07 2021.